

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	SECTION:	CPE31S2
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ACTIVITY NO. 10.1 Install, Configure and Manage Enterprise Log Monitoring via Ansible

1. PROCEDURE OUTPUT

I. Import ElasticSearch GPG Key

1. Create the roles directory for ElasticSearch with `mkdir -p roles/ElasticSearch/tasks`
2. Create the role playbook with `nano roles/ElasticSearch/tasks/main.yml` and create the play for gpg key installation.

```

qimagalan-workstation@workstation: ~/... x qimagalan-workstation@workstation: ~/E... x qimagalan-workstation@workstation: ~/E...
GNU nano 7.2 roles/ElasticSearch/tasks/main.yml

- name: Create directory for installation
  ansible.builtin.file:
    path: "{{ elastic_dir }}"
    state: directory

- name: Download the signing key
  become: true
  ansible.builtin.get_url:
    url: "{{ signkey_url }}"
    dest: "{{ elastic_dir }}"
  when: ansible_distribution == "Ubuntu"

- name: Extract gpg key
  become: true
  ansible.builtin.command:
    cmd: gpg --dearmor --output /usr/share/keyrings/elasticsearch-keyring.gpg "{{ elastic_dir }}"
    creates: /usr/share/keyrings/elasticsearch-keyring.gpg
  when: ansible_distribution == "Ubuntu"

- name: Import Signing Key for CentOS
  become: true
  ansible.builtin.rpm_key:
    key: "{{ signkey_url }}"
    state: present
  when: ansible_distribution == "CentOS"

```

This creates the directory we need to work with the ElasticStack plus it downloads the signing key as preparatory step as recommended in the documentation:

<https://www.elastic.co/docs/deploy-manage/deploy/self-managed/install-elasticsearch-with-debian-package#deb-key>

3. Create the following variables inside `roles/ElasticSearch/vars/main.yml`

```
qimagalan-workstation@workstation: ~/... x qimagalan-workstation@workstation: ~/E... x qimagalan-workstation@workstation: ~/E... x
GNU nano 7.2 roles/ElasticSearch/vars/main.yml
---
signkey_url: "https://artifacts.elastic.co/GPG-KEY-elasticsearch"
debpkg_link: "https://artifacts.elastic.co/downloads/elasticsearch/elasticsearch-9.5.4-amd64.deb"
debpkg_chksun: "https://artifacts.elastic.co/downloads/elasticsearch/elasticsearch-9.5.4-amd64.deb.sha512"
rpmkg_link: "https://artifacts.elastic.co/downloads/elasticsearch/elasticsearch-9.5.4-x86_64.rpm"
rpmkg_chksun: "https://artifacts.elastic.co/downloads/elasticsearch/elasticsearch-9.5.4-x86_64.rpm.sha512"
elastic_dir: "{{ ansible_env.HOME }}/ElasticStack"
```

4. Run the playbook to install the gpg keys for Ubuntu and CentOS

```
qimagalan-workstation@workstation:~/ElasticStackInstaller_Galan$ ansible-playbook install_elasticstack.yml -i
inventory.yml -K
BECOME password:

PLAY [web_servers,file_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.114]
ok: [192.168.56.113]

TASK [ElasticSearch : Create directory for installation] *****
ok: [192.168.56.113]
ok: [192.168.56.114]

TASK [ElasticSearch : Download the signing key] *****
skipping: [192.168.56.114]
ok: [192.168.56.113]

TASK [ElasticSearch : Extract gpg key] *****
skipping: [192.168.56.114]
ok: [192.168.56.113]

TASK [ElasticSearch : Import Signing Key for CentOS] *****
skipping: [192.168.56.113]
ok: [192.168.56.114]

PLAY RECAP *****
192.168.56.113      : ok=4    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    igno
red=0
192.168.56.114      : ok=3    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    igno
red=0
```

```
qimagalan-workstation@server1:~$ ls ~/ElasticStack
GPG-KEY-elasticsearch
qimagalan-workstation@server1:~$
```

II. Download from the debian and rpm links

1. Modify the playbook and the play that will install the deb and rpm packages

```
- name: Download Elastic Search deb pkg
  ansible.builtin.get_url:
    url: "{{ debpkg_link }}"
    dest: "{{ elastic_dir }}"
    checksum: "sha512:{{ debpkg_chksum }}"
    when: ansible_distribution == "Ubuntu"

- name: Download Elastic Search rpm pkg
  ansible.builtin.get_url:
    url: "{{ rpmpkg_link }}"
    dest: "{{ elastic_dir }}"
    checksum: "sha512:{{ rpmpkg_chksum }}"
    when: ansible_distribution == "CentOS"
```

2. Run the playbook and check results

```
TASK [ElasticSearch : Extract gpg key] *****
skipping: [192.168.56.114]
ok: [192.168.56.113]

TASK [ElasticSearch : Import Signing Key for CentOS] *****
skipping: [192.168.56.113]
ok: [192.168.56.114]

TASK [ElasticSearch : Download Elastic Search deb pkg] *****
skipping: [192.168.56.114]
changed: [192.168.56.113]

TASK [ElasticSearch : Download Elastic Search rpm pkg] *****
skipping: [192.168.56.113]
changed: [192.168.56.114]

PLAY RECAP *****
192.168.56.113      : ok=5    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    igno
red=0
192.168.56.114      : ok=4    changed=1    unreachable=0    failed=0    skipped=3    rescued=0    igno
red=0

qimagalan-workstation@workstation:~/ElasticStackInstaller_Galan$
```

```
qimagalan-workstation@server1:~$ ls ~/ElasticStack
elasticsearch-9.5.4-amd64.deb  GPG-KEY-elasticsearch
qimagalan-workstation@server1:~$
```

III. Install the packages downloaded

1. Add these variables to the roles/ElasticSearch/vars/main.yml

```
qimagalan@localhost:~ x qimagalan-workstation@workstation: ~/E... x qimagalan-workstation@workstation: ~/E... x
GNU nano 7.2 roles/ElasticSearch/vars/main.yml *
---
signkey_url: "https://artifacts.elastic.co/GPG-KEY-elasticsearch"
debpkg_link: "https://artifacts.elastic.co/downloads/elasticsearch/elasticsearch-9.5.4-amd64.deb"
debpkg_chksum: "https://artifacts.elastic.co/downloads/elasticsearch/elasticsearch-9.5.4-amd64.deb.sha512"
rpmkg_link: "https://artifacts.elastic.co/downloads/elasticsearch/elasticsearch-9.5.4-x86_64.rpm"
rpmkg_chksum: "https://artifacts.elastic.co/downloads/elasticsearch/elasticsearch-9.5.4-x86_64.rpm.sha512"
elastic_dir: "{{ ansible_env.HOME }}/ElasticStack"

deb_elastic: "{{ elastic_dir }}/elasticsearch-9.5.4-amd64.deb"
rpm_elastic: "{{ elastic_dir }}/elasticsearch-9.5.4-x86_64.rpm"
```

2. Add this to the bottom of the playbook to install from packages

```
- name: Install from deb pkg
  tags: install
  become: true
  ansible.builtin.apt:
    deb: "{{ deb_elastic }}"
  when: ansible_distribution == "Ubuntu"

- name: Install from rpm pkg
  tags: install
  become: true
  ansible.builtin.dnf:
    name: "{{ rpm_elastic }}"
    state: present
  when: ansible_distribution == "CentOS"
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo
 ^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line M-E Redo

3. Run the playbook and check results

```
qimagalan-workstation@workstation: ~/... x qimagalan-workstation@workstation: ~/E... x qimagalan-workstation@workstation: ~/E... x
qimagalan-workstation@workstation:~/ElasticStackInstaller_Galan$ ansible-playbook install_elasticstack.yml -i
inventory.yml -K --tags "install"
BECOME password:

PLAY [web_servers,file_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.113]
ok: [192.168.56.114]

TASK [ElasticSearch : Install from deb pkg] *****
skipping: [192.168.56.114]
changed: [192.168.56.113]

TASK [ElasticSearch : Install from rpm pkg] *****
skipping: [192.168.56.113]
changed: [192.168.56.114]

PLAY RECAP *****
192.168.56.113      : ok=2    changed=1    unreachable=0    failed=0    skipped=1    rescued=0    igno
red=0
192.168.56.114      : ok=2    changed=1    unreachable=0    failed=0    skipped=1    rescued=0    igno
red=0
qimagalan-workstation@workstation:~/ElasticStackInstaller_Galan$
```

```
qimagalan@localhost:~ x qimagalan-workstation@workstation: ~/E... x qimagalan-workstation@workstation: ~/E... x
qimagalan-workstation@workstation:~/ElasticStackInstaller_Galan$ ssh qimagalan@Galan_CentOS
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Tue Sep 29 17:11:36 2026 from 192.168.56.1
[qimagalan@localhost ~]$ sudo systemctl status elasticsearch
[sudo] password for qimagalan:
○ elasticsearch.service - Elasticsearch
   Loaded: loaded (/usr/lib/systemd/system/elasticsearch.service; disabled; preset: disabled)
   Active: inactive (dead)
     Docs: https://www.elastic.co
[qimagalan@localhost ~]$
```

IV. Setup Nodes for connection

1. Add another play at the bottom to place basic configuration on the hosts

```
- name: Configure the elasticsearch.yml for both hosts
tags: config
become: true
ansible.builtin.blockinfile:
  path: /etc/elasticsearch/elasticsearch.yml
  block: |
    cluster.name: my-cluster
    network.host: 0.0.0.0
    transport.host: 0.0.0.0
  marker: "# {mark} ANSIBLE MANAGED ELASTICSEARCH CONFIG"
```

^{^G} Help ^{^O} Write Out ^{^W} Where Is ^{^K} Cut ^{^T} Execute
^{^X} Exit ^{^R} Read File ^{^_} Replace ^{^U} Paste ^{^J} Justify

2. Run and check if configurations reflected

```
PLAY [web_servers,file_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.113]
ok: [192.168.56.114]

TASK [ElasticSearch : Configure the elasticsearch.yml for both hosts] *****
changed: [192.168.56.113]
changed: [192.168.56.114]

PLAY RECAP *****
192.168.56.113      : ok=2    changed=1    unreachable=0    failed=0    skipped=
0      rescued=0    ignored=0
192.168.56.114      : ok=2    changed=1    unreachable=0    failed=0    skipped=
0      rescued=0    ignored=0
```

qimagalan-workstation@workstation:~/ElasticStackInstaller_Galan\$

#transport.host: 0.0.0.0

#----- END SECURITY AUTO CONFIGURATION -----

BEGIN ANSIBLE MANAGED ELASTICSEARCH CONFIG

cluster.name: my-cluster

network.host: 0.0.0.0

transport.host: 0.0.0.0

END ANSIBLE MANAGED ELASTICSEARCH CONFIG

qimagalan-workstation@server1:~\$

3. Add the play to start and enable the elasticsearch service

```
- name: Start the elasticsearch service
  tags: start
  become: true
  ansible.builtin.service:
    name: elasticsearch
    state: started
    enabled: true
```

[Wrote 64 lines]

^G Help

^O Write Out

^W Where Is

^K Cut

^T Exe

^X Exit

^R Read File

^_ Replace

^U Paste

^J Jus

4. Run playbook to start service first

```
PLAY [web_servers,file_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.113]
ok: [192.168.56.114]

TASK [ElasticSearch : Start the elasticsearch service] *****
changed: [192.168.56.114]
changed: [192.168.56.113]

PLAY RECAP *****
192.168.56.113      : ok=2    changed=1    unreachable=0    failed=0    skipped=
0    rescued=0    ignored=0
192.168.56.114      : ok=2    changed=1    unreachable=0    failed=0    skipped=
0    rescued=0    ignored=0

qimagalan-workstation@workstation:~/ElasticStackInstaller_Galan$
```

```
qimagalan-workstation@server1:~$ sudo systemctl status elasticsearch
[sudo] password for qimagalan-workstation:
● elasticsearch.service - Elasticsearch
   Loaded: loaded (/usr/lib/systemd/system/elasticsearch.service; enabled; preset: e>
   Active: active (running) since Tue 2026-09-29 17:21:59 PST; 1min 8s ago
     Docs: https://www.elastic.co
   Main PID: 11271 (server-launcher)
    Tasks: 104 (limit: 1432)
  Memory: 1.2G (peak: 1.3G swap: 446.8M swap peak: 446.8M)
     CPU: 3min 5.431s
   CGroup: /system.slice/elasticsearch.service
           └─11271 /usr/share/elasticsearch/lib/tools/server-launcher/server-launcher>
             └─11327 /usr/share/elasticsearch/jdk/bin/java -Des.networkaddress.cache.t>
               └─11354 /usr/share/elasticsearch/modules/x-pack-ml/platform/linux-x86_64/>

Sep 29 17:19:49 server1 systemd[1]: Starting elasticsearch.service - Elasticsearch...
Sep 29 17:20:18 server1 systemd-entrypoint[11271]: WARNING: Unknown module: org.apache>
Sep 29 17:21:54 server1 systemd-entrypoint[11271]: WARNING: Automatically detected m>
```

5. Reload the system daemon as well by adding a new play

```
- name: Reload systemd daemon
  tags: reload
  become: true
  ansible.builtin.systemd:
    daemon_reload: true
```

```
qimagalan-workstation@serv... x qimagalan-workstation@work... x qimagalan-workstation@work... x
ll_elasticstack.yml -i inventory.yml -K --tags "reload"
BECOME password:

PLAY [web_servers,file_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.113]
ok: [192.168.56.114]

TASK [ElasticSearch : Reload systemd daemon] *****
ok: [192.168.56.113]
ok: [192.168.56.114]

PLAY RECAP *****
192.168.56.113 : ok=2 changed=0 unreachable=0 failed=0 skipped=
0 rescued=0 ignored=0
192.168.56.114 : ok=2 changed=0 unreachable=0 failed=0 skipped=
0 rescued=0 ignored=0
```

6. Then reset the password by doing:
/usr/share/elasticsearch/binelasticsearch-reset-password -u elastic
7. Take note of the password as it will be used to verify the installation of elasticsearch

```
qimagalan@localhost:/usr/share/elasticsearch/bin x qimagalan-workstation@v
[qimagalan@localhost bin]$ sudo ./elasticsearch-reset-password -u elastic
[sudo] password for qimagalan:
This tool will reset the password of the [elastic] user to an autogenerated value.
The password will be printed in the console.
Please confirm that you would like to continue [y/N]y

Password for the [elastic] user successfully reset.
New value: GPWqTa9+v9ag*z7sMTsD
[qimagalan@localhost bin]$
```

V. Verify Installation of Elastic Search

1. Run `sudo curl --cacert /etc/elasticsearch/certs/http_ca.crt -u elastic https://localhost:9200`. Then type the password from the reset earlier.

```
qimagalan-workstation@server1:~$ sudo curl --cacert /etc/elasticsearch/certs/http_ca.crt \
-u elastic https://localhost:9200
Enter host password for user 'elastic':
{
  "name" : "server1",
  "cluster_name" : "my-cluster",
  "cluster_uuid" : "7CVA-1HmTaG2Tu204WPK1Q",
  "version" : {
    "number" : "9.5.4",
    "build_flavor" : "default",
    "build_type" : "deb",
    "build_hash" : "9170df19cae1adb107b7b489b4d82dec66d7a337",
    "build_date" : "2026-09-09T22:42:53.976833287Z",
    "build_snapshot" : false,
    "lucene_version" : "10.5.1",
    "minimum_wire_compatibility_version" : "8.19.0",
    "minimum_index_compatibility_version" : "8.0.0"
  },
  "tagline" : "You Know, for Search"
}
qimagalan-workstation@server1:~$
```

```
qimagalan@localhost:/usr/share/elasticsearch/bin x qimagalan-workstation
[qimagalan@localhost bin]$ sudo curl --cacert /etc/elasticsearch/certs/http_ca.crt -
Enter host password for user 'elastic':
{
  "name" : "localhost.localdomain",
  "cluster_name" : "my-cluster",
  "cluster_uuid" : "q4gPgla0TfGFkU0i86W4Mw",
  "version" : {
    "number" : "9.5.4",
    "build_flavor" : "default",
    "build_type" : "rpm",
    "build_hash" : "9170df19cae1adb107b7b489b4d82dec66d7a337",
    "build_date" : "2026-09-09T22:42:53.976833287Z",
    "build_snapshot" : false,
    "lucene_version" : "10.5.1",
    "minimum_wire_compatibility_version" : "8.19.0",
    "minimum_index_compatibility_version" : "8.0.0"
  },
  "tagline" : "You Know, for Search"
}
[qimagalan@localhost bin]$
```

VI. Install Kibana through deb and rpm package

1. Create the role for the Kibana installation: `mkdir -p roles/Kibana/tasks`
2. Create the roles/Kibana/vars/main.yml file and place the following variables

```
qimagalan@localhost:~ x qimagalan-workstation@server1: ~ x qimagalan-workstation@workstation: ~/Ela...
GNU nano 7.2 roles/Kibana/vars/main.yml
--
debpkg_link: "https://artifacts.elastic.co/downloads/kibana/kibana-9.5.4-amd64.deb"
debpkg_chksum: "https://artifacts.elastic.co/downloads/kibana/kibana-9.5.4-amd64.deb.sha512"
rpmpkg_link: "https://artifacts.elastic.co/downloads/kibana/kibana-9.5.4-x86_64.rpm"
rpmpkg_chksum: "https://artifacts.elastic.co/downloads/kibana/kibana-9.5.4-x86_64.rpm.sha512"
elastic_dir: "{{ ansible_env.HOME }}/ElasticStack"

deb_elastic: "{{ elastic_dir }}/kibana-9.5.4-amd64.deb"
rpm_elastic: "{{ elastic_dir }}/kibana-9.5.4-x86_64.rpm"
```

3. Create the playbook to install Kibana *roles/Kibana/tasks/main.yml*

```
qimagalan@l... x qimagalan-w... x qimagalan-w... x qimagalan-w... x
GNU nano 7.2 roles/Kibana/tasks/main.yml
--
- name: Install required packages
  become: true
  ansible.builtin.apt:
    name:
      - apt-transport-https
    state: present
  when: ansible_distribution == "Ubuntu"

- name: Download Kibana deb pkg
  ansible.builtin.get_url:
    url: "{{ debpkg_link }}"
    dest: "{{ elastic_dir }}"
    checksum: "sha512:{{ debpkg_chksum }}"
  when: ansible_distribution == "Ubuntu"

- name: Download Kibana rpm pkg
  ansible.builtin.get_url:
    url: "{{ rpmpkg_link }}"
    dest: "{{ elastic_dir }}"
    checksum: "sha512:{{ rpmpkg_chksum }}"
  when: ansible_distribution == "CentOS"
```

```
- name: Install from deb pkg
tags: install
become: true
ansible.builtin.apt:
  deb: "{{ deb_elastic }}"
when: ansible_distribution == "Ubuntu"

- name: Install from rpm pkg
tags: install
become: true
ansible.builtin.dnf:
  name: "{{ rpm_elastic }}"
  state: present
when: ansible_distribution == "CentOS"
```

4. Run the playbook and verify results

```
qimagalan-workstation@workstation:~/ElasticStackInstaller_Galan$ ansible-playbook ins
lasticstack.yml -i inventory.yml -K --tags "kibana"
BECOME password:

PLAY [web_servers,file_servers] *****

PLAY [web_servers,file_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.113]
ok: [192.168.56.114]

TASK [Kibana : Install required packages] *****
skipping: [192.168.56.114]
ok: [192.168.56.113]

TASK [Kibana : Download Kibana deb pkg] *****
skipping: [192.168.56.114]
changed: [192.168.56.113]

TASK [Kibana : Download Kibana rpm pkg] *****
skipping: [192.168.56.113]
ok: [192.168.56.114]

TASK [Kibana : Install from deb pkg] *****
skipping: [192.168.56.114]
changed: [192.168.56.113]

TASK [Kibana : Install from rpm pkg] *****
skipping: [192.168.56.113]
ok: [192.168.56.114]

PLAY RECAP *****
192.168.56.113      : ok=4    changed=2    unreachable=0    failed=0    skippe
rescued=0    ignored=0
```

```
qimagalan@localhost:~ x qimagalan-workstation@server1: ~
[qimagalan@localhost ~]$ ls ~/ElasticStack
elasticsearch-9.5.4-x86_64.rpm kibana-9.5.4-x86_64.rpm
[qimagalan@localhost ~]$
```

5. Generate a kibana enrollment token by adding this play

```
[gimigalan@localhost ElasticStack]$
```

```
qimagalan-workstation@server1:~$ cat ~/ElasticStack/kibanatoken  
"eyJ2ZXI0eSI0LjE0LjAiLCJhZHI0LSiMTAUMC4yLjE0IjkyMDAiXSwiZm9ycjoOWJmNzU0Yzc2MzAxMTliMzM4OGM3MWJkODNiNWZjODk1NDVlbnEzOGM0MzRkZDcxMzljMjY2OWM3Y2Q3NGM3OSIsIntleSI6ImRPbm43YUFCQnJBV0pTTjJlak5ROkRWRFo1YVF5SWFlqbWljTwd3aWV6V1EifQ=="
```

7. Start and Enable the kibana service by adding this in the Kibana roles tasks

```
- name: Start and Enable Kibana Service
  ansible.builtin.service:
    name: kibana
    state: started
    enabled: true
  become: true

- name: Reload systemd daemon
  tags: reload
  become: true
  ansible.builtin.systemd:
    daemon_reload: true
```

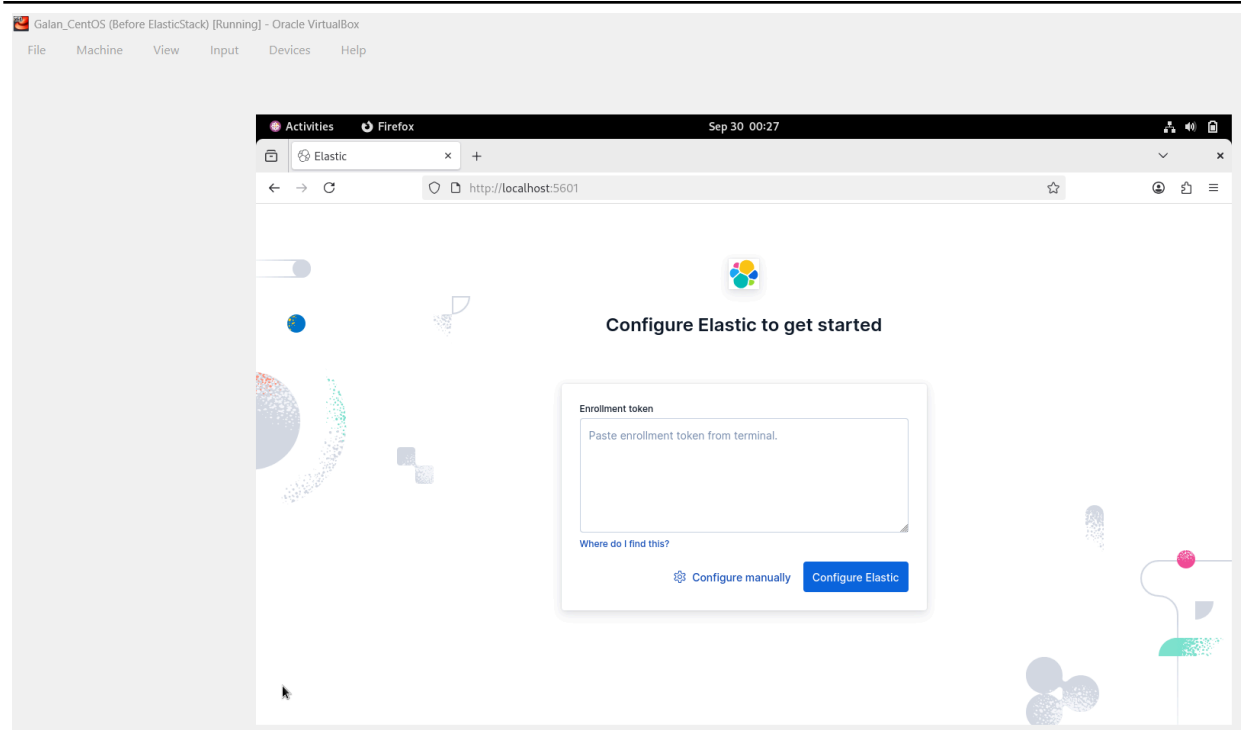
```
[qimagalan@localhost ElasticStack]$ sudo systemctl status kibana
[sudo] password for qimagalan:
● kibana.service - Kibana
   Loaded: loaded (/usr/lib/systemd/system/kibana.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-09-29 16:47:31 PST; 59s ago
     Docs: https://www.elastic.co
    Main PID: 17916 (MainThread)
      Tasks: 11 (limit: 10500)
     Memory: 487.6M (peak: 487.7M)
        CPU: 28.668s
    CGroup: /system.slice/kibana.service
            └─17916 /usr/share/kibana/bin/../node/default/bin/node /usr/share/kibana/bin/>
```

```
Sep 29 16:47:31 localhost.localdomain systemd[1]: Started Kibana.
Sep 29 16:47:57 localhost.localdomain kibana[17916]: {"log.level":"info","@timestamp":"202>
Sep 29 16:47:57 localhost.localdomain kibana[17916]: Native global console methods have be>
Sep 29 16:48:11 localhost.localdomain kibana[17916]: [2026-09-29T16:48:11.356+08:00][INFO >
Sep 29 16:48:11 localhost.localdomain kibana[17916]: [2026-09-29T16:48:11.509+08:00][INFO >
```

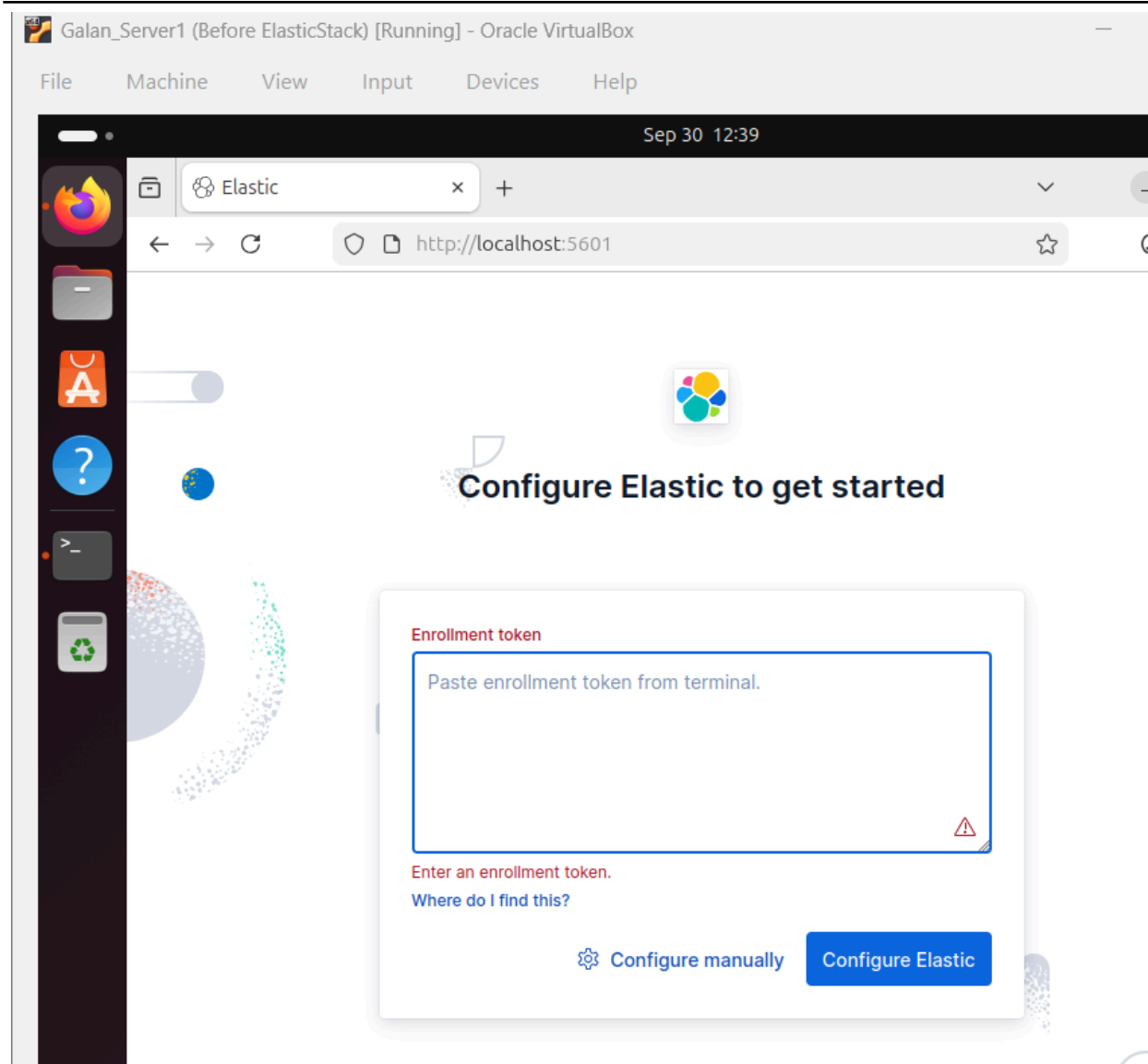
lines 1-16/16 (END)

8. Verify the installation of kibana in port 5601

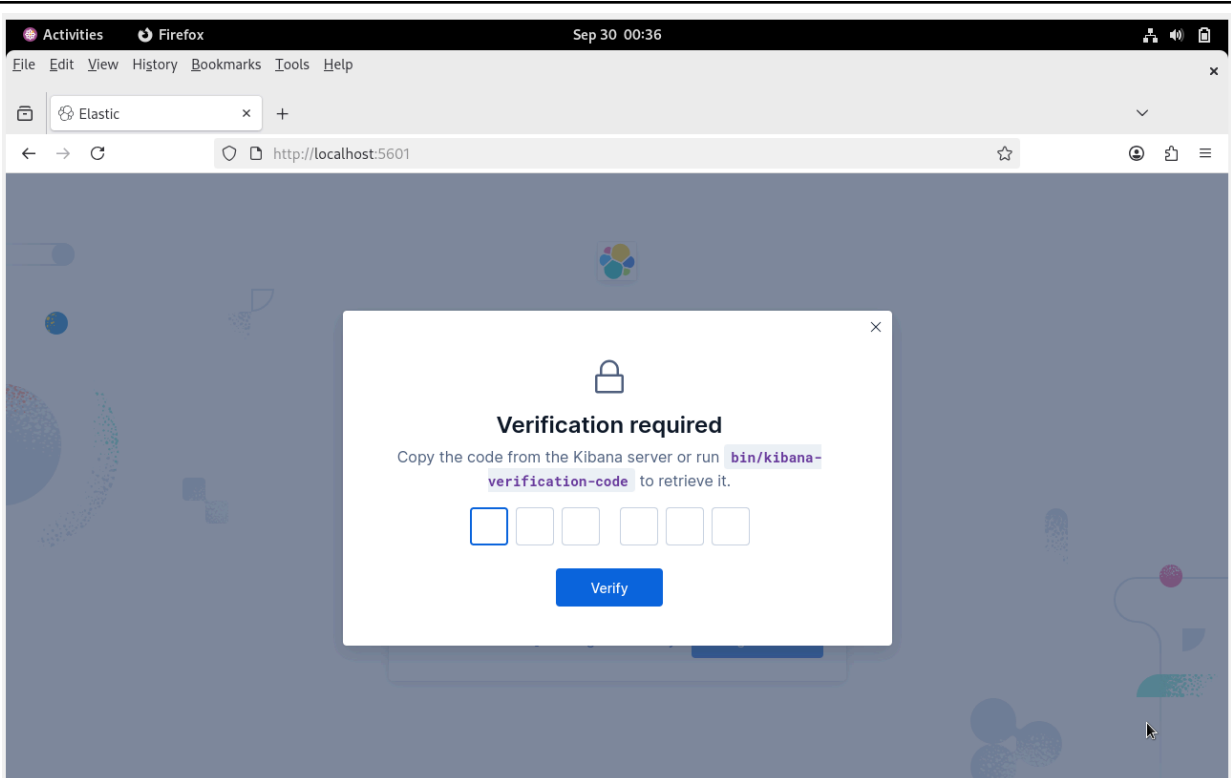
CentOS:



Ubuntu:



9. Use the generated enrollment token from the play in the former steps and paste in the box



10. The code is viewable via `sudo journalctl -u kibana`

```
qimagalan@localhost:~ — sudo journalctl -u kibana
Sep 30 00:20:04 localhost.localdomain systemd[1]: Started Kibana.
Sep 30 00:21:10 localhost.localdomain kibana[1127]: {"log.level":"info","@timestamp":"2026-09-2>
Sep 30 00:21:12 localhost.localdomain kibana[1127]: Native global console methods have been ove>
Sep 30 00:21:26 localhost.localdomain kibana[1127]: [2026-09-30T00:21:26.418+08:00][INFO ][root>
Sep 30 00:21:26 localhost.localdomain kibana[1127]: [2026-09-30T00:21:26.689+08:00][INFO ][node>
Sep 30 00:22:24 localhost.localdomain kibana[1127]: [2026-09-30T00:22:23.990+08:00][INFO ][plug>
Sep 30 00:22:24 localhost.localdomain kibana[1127]: [2026-09-30T00:22:24.744+08:00][INFO ][http>
Sep 30 00:22:26 localhost.localdomain kibana[1127]: [2026-09-30T00:22:26.124+08:00][INFO ][plug>
Sep 30 00:22:26 localhost.localdomain kibana[1127]: [2026-09-30T00:22:26.292+08:00][INFO ][preb>
Sep 30 00:22:26 localhost.localdomain kibana[1127]: [2026-09-30T00:22:26.495+08:00][INFO ][root>
Sep 30 00:22:35 localhost.localdomain kibana[1127]: i Kibana has not been configured.
Sep 30 00:22:35 localhost.localdomain kibana[1127]: Go to http://localhost:5601/?code=533312 to>
Sep 30 00:35:43 localhost.localdomain kibana[1127]: Your verification code is: 533 312
Sep 30 00:38:20 localhost.localdomain systemd[1]: Stopping Kibana...
Sep 30 00:38:20 localhost.localdomain kibana[1127]: [2026-09-30T00:38:20.088+08:00][INFO ][root>
Sep 30 00:38:20 localhost.localdomain kibana[1127]: [2026-09-30T00:38:20.325+08:00][INFO ][root>
Sep 30 00:38:21 localhost.localdomain kibana[1127]: [2026-09-30T00:38:21.283+08:00][INFO ][plug>
Sep 30 00:38:21 localhost.localdomain kibana[1127]: [2026-09-30T00:38:21.401+08:00][INFO ][plug>
Sep 30 00:38:22 localhost.localdomain systemd[1]: kibana.service: Deactivated successfully.
Sep 30 00:38:22 localhost.localdomain systemd[1]: Stopped Kibana.
Sep 30 00:38:22 localhost.localdomain systemd[1]: kibana.service: Consumed 1min 22.732s CPU tim>
lines 1-21
```

verification-code to retrieve it.

5 3 3 3 1 2

Verify

VII. Install Logstash

1. Create a role for logstash through `mkdir -p roles/Logstash/tasks`
2. Create the Logstash play for installing logstash from apt and yum repositories through `nano roles/Logstash/tasks/main.yml`

```
qimagalan-workstation@workstation: ~/ElasticStackInstaller_Galan
qimagalan-workstation@workstation: ~/ElasticStackInstaller_Galan
GNU nano 7.2 roles/Logstash/tasks/main.yml *
- name: Add the apt repository to sources list
  become: true
  ansible.builtin.apt_repository:
    repo: "deb [signed-by=/usr/share/keyrings/elastic-keyring.gpg] https://artifacts.elastic.co/packages/9.x/apt stable main"
    filename: elastic-9.x
    state: present
    update_cache: true #refresh the repository list after adding
  when: ansible_distribution == "Ubuntu"

- name: Add the yum repository source
  become: true
  ansible.builtin.yum_repository:
    name: elastic-9.x
    description: Elastic repository for 9.x packages
    baseurl: https://artifacts.elastic.co/packages/9.x/yum
    gpgcheck: true
    gpgkey: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    enabled: true
    file: elastic-9.x
```

```
- name: Install the logstash from repository source
  become: true
  ansible.builtin.package:
    name: logstash
    state: present
```

3. Run playbook and check if Logstash is installed

```
qimagalan-workstation@workstation: ~/ElasticStackInstaller_Galan
qimagalan-workstation@workstation: ~/ElasticSta... x qimagalan-workstation@workstation: ~/ElasticSta... x qimagalan-workstation@server1: ~
ok: [192.168.56.114]
ok: [192.168.56.113]

TASK [Logstash : Download the signing key] *****
skipping: [192.168.56.114]
ok: [192.168.56.113]

TASK [Logstash : Extract gpg key] *****
skipping: [192.168.56.114]
ok: [192.168.56.113]

TASK [Logstash : Add the apt repository to sources list] *****
skipping: [192.168.56.114]
ok: [192.168.56.113]

TASK [Logstash : Add the yum repository source] *****
skipping: [192.168.56.113]
ok: [192.168.56.114]

TASK [Logstash : Refresh repository list before installing logstash] *****
changed: [192.168.56.113]
ok: [192.168.56.114]

TASK [Logstash : Install the logstash from repository source] *****
ok: [192.168.56.113]
ok: [192.168.56.114]

PLAY RECAP *****
192.168.56.113 : ok=6 changed=1 unreachable=0 failed=0 skipped=1 rescued=0 ignored=0
192.168.56.114 : ok=4 changed=0 unreachable=0 failed=0 skipped=3 rescued=0 ignored=0
```

```
qimagalan-workstation@workstation: ~/ElasticSta... x qimagalan-workstation@workstation: ~/ElasticSta... x
qimagalan-workstation@server1:~$ sudo systemctl status logstash
[sudo] password for qimagalan-workstation:
○ logstash.service - logstash
   Loaded: loaded (/usr/lib/systemd/system/logstash.service; disabled; preset: enabled)
   Active: inactive (dead)
```

```
qimagalan-workstation@workstation:~/ElasticStackInstaller_Galan$ ssh qimagalan@Galan_CentOS
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Wed Sep 30 10:46:12 2026 from 192.168.56.1
[qimagalan@localhost ~]$ sudo systemctl status logstash
[sudo] password for qimagalan:
○ logstash.service - logstash
   Loaded: loaded (/usr/lib/systemd/system/logstash.service; disabled; preset: disabled)
   Active: inactive (dead)
[qimagalan@localhost ~]$
```

4. Create a play to start logstash

```
- name: Start and Enable Logstash
  become: true
  ansible.builtin.service:
    name: logstash
    state: started
    enabled: true
```

5. Run the playbook again to start logstash

```
TASK [Logstash : Start and Enable Logstash] *****
changed: [192.168.56.113]
changed: [192.168.56.114]

PLAY RECAP *****
192.168.56.113      : ok=7    changed=2    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.114      : ok=5    changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0

qinagalan-workstation@workstation:~/ElasticStackInstaller_Galan$
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

What are the benefits of having a log monitoring tool?

Log monitoring tools are not an optional part of system administration, but it is an imperative part of all processes and services running on a machine. Having server automation does not automatically mean letting the application or service run completely on its own. There will be scenarios where errors may suddenly appear and without having a log monitoring tool, it becomes impossible to track the causes of failure. Every debugging process also starts with identifying the errors that occurred within the process' runtime. The log monitoring tools also provide the centralized view of the activities of the system. Regardless of the specific event that occurred whether an error or authentication problems, these are only visible due to logs

Conclusion

I have learned important concepts in managing installation of a multi-service application stack such as the ELK. Application stack shares a common GPG key to sign and authenticate packages added to the repository sources list. Without these the installation from deb and rpm packages will fail due to authentication. Another useful feature of ansible I have learned is the register variable that we can use to suppress standard output and store the value of output for usage of other plays. This is especially useful for storing the enrollment tokens that we can use to provide connections for Kibana and nodes in Elastic Stack.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of

concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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